

Human Biology

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Lipids

- ❖ Lipids are organic compounds that contain hydrogen, carbon, and oxygen atoms, which form the framework for the structure and function of living cells.
- ❖ Lipids are a group of organic compounds that include fats, oils, waxes, steroids, and phospholipids. They are essential for life on Earth and perform many important roles in the body, including:
 - Cell membranes: Lipids are a major structural component of cell membranes, helping control what goes in and out of cells.
 - Energy storage: Lipids help store energy for the future.
 - Hormone production: Lipids are precursors for hormones like estrogen and testosterone.
 - Vitamin absorption: Lipids help the body absorb vitamins.
 - Chemical messengers: Lipids act as chemical messengers in the body.
 - Protection: Waxes help protect plants from dehydration and animals' feathers or fur from water damage.

Lipids are insoluble in water but soluble in non-polar solvents. They are made up of fatty acids with different chain lengths and degrees of saturation. Saturated fats have no double bonds between carbon atoms, while unsaturated fats have one or more double bonds.

Classification Of Lipids

- Lipids are classified in a number of ways, including by their physical properties, polarity, and importance to the human body. Some types of lipids include:
- Fatty acids: The simplest type of lipid, fatty acids are made up of long chains of carbon atoms with hydrogen atoms attached. They are divided into two groups: saturated and unsaturated.
- Phospholipids: A key component of the cell membrane, phospholipids contain two fatty acids attached to glycerol, with a phosphate group at the third glycerol binding site.
- Glycolipids: These lipids contain a glycosidic carbohydrate and help provide structural stability to the cell membrane. They also play a role in cell recognition.
- Nonsaponifiable lipids: These lipids do not undergo hydrolysis under alkaline conditions. They include steroids, terpenes, pheromones, isoprenoids, eicosanoids, and fat-soluble vitamins.
- Glycerophospholipids: These lipids are a key part of a cell's lipid layering and help with cell metabolism. Alterations in glycerophospholipids in the brain can cause neurological conditions.
- Waxes: These esters are made from long-chain fatty acids and long-chain alcohols.
- Lipoproteins: These compounds are made up of a lipid and a protein and play a major role in lipid and cholesterol transport and metabolism.

Fat

- ❖ Fats are nutrients in food that the body uses to build cell membranes, nerve tissue (including the brain), and hormones. The body also uses fat as fuel. If fats eaten aren't burned as energy or used as building blocks, they're stored by the body in fat cells.
- ❖ Fats are classified based on the degree of saturation or unsaturation in their carbon chains, or the number of double bonds they have:
 - ❖ Saturated fats: Have no double bonds and are solid at room temperature
 - ❖ Monounsaturated fats: Have one double bond
 - ❖ Polyunsaturated fats: Have two or more double bonds
 - ❖ Trans fats: A type of unsaturated fatty acid that can be naturally occurring or industrially produced
 - ❖ Omega-3 fatty acids: A type of polyunsaturated fat that the body can't make and must be obtained from diet

Difference between Fats and Lipids

Lipids	Fats
These are not soluble in water, but in organic solvents, they are soluble.	These are oily substances found beneath the skin and surrounding the organs.
These are a varied group of biomolecules.	These are a kind of lipids.
These are solids as well as liquids.	These are solids only.
These are structural components and energy molecules in the body.	This serves as a source of energy.
There are three types – steroids, triglycerides, and phospholipids.	These are of two types – saturated and unsaturated fats.

Lipids: These are chemical compounds in your body that help with some of its functions.

Fats: These are nutrients in food that the body uses to build cell membranes, nerve tissue (including the brain), and hormones. It is also called Fatty Acids or Lipids. Fats in our body are made up of three molecules joined together. These three molecule structure is called Triglyceride

Sources of Fats and Oils

- ❖ Oils are commonly obtained from oil producing plants like corn, peanuts, soya beans, vegetable seeds, olives, palm kernels, whilst fats are more commonly obtained from animal sources like milk (butter), beef (tallow), pork (lard), etc. Fats and oils can also be obtained from marine sources like fish.

Types of Vitamins

- ❖ Vitamins are nutrients that the body needs in small amounts to function and stay healthy. They are organic substances that come from plants or animals, and are essential for normal cell function, growth, and development.
- ❖ Based on the solubility, Vitamins have been classified into two different groups:
 - ✓ Fat-Soluble Vitamins.
 - ✓ Water-Soluble Vitamins."

Sources of Vitamins

- ❖ "The human body is so designed that it takes what it needs from the food we eat and then it passes out waste as excreta. These organic substances are abundantly found in both plants and animals source and play a vital role in both growth and development and optimal health. Listed below are the different types of vitamins along with their sources. The best sources of fat-soluble vitamins include:
- Vitamin A: Found in potato, carrots, pumpkins, spinach, beef and eggs.
 - Vitamin D: Found in fortified milk and other dairy products.
 - Vitamin E: Found in fortified cereals, leafy green vegetables, seeds, and nuts.
 - Vitamin K: Found in dark green leafy vegetables and in turnip or beet green.

Other Sources of Vitamins

❖ Other sources of Vitamin B and C include:

- Vitamin B1 or Thiamin: Found in pork chops, ham, enriched grains and seeds.
- Vitamin B2 or Riboflavin: Found in whole grains, enriched grains and dairy products.
- Vitamin B3 or Niacin: Found in mushrooms, fish, poultry, and whole grains.
- Vitamin B5 or Pantothenic Acid: Found in chicken, broccoli, legumes and whole grains.
- Vitamin B6 or Pyridoxine: Found in fortified cereals and soy products.
- Vitamin B7 or Biotin: Found in many fruits like fruits and meats.
- Vitamin B9 or Folic Acid: Found in leafy vegetables.
- Vitamin B12: Found in fish, poultry, meat and dairy products.
- Vitamin C: Found in citrus fruits and juices, such as oranges and grapefruits."

Thank You